

Shafikul Islam Marwan

marwanthe0@gmail.com | +8801647040299 | Sylhet, Bangladesh | [linkedin.com/in/marwanahmed27](https://www.linkedin.com/in/marwanahmed27)
github.com/Marwanthe0 | codeforces.com/profile/TheHackerBd | leetcode.com/u/marwanthe0

EDUCATION

Bachelor Of Science in Computer Science and Engineering

08/2023 – Present | Jashore, Bangladesh

Jashore University of Science and Technology

CGPA: 3.67/4.00 (till 6th semester)

SKILLS

- **Programming Languages:** C++, Python, C#, SQL
- **Embedded & Hardware:** ESP32, TinyML, Arduino/C++, 1-Wire Protocol, ADC Interfacing, Relay Actuation, PWM
- **ML/Data Science:** Pandas, NumPy, Matplotlib, Seaborn, scikit-learn
- **Frameworks:** STL, FastAPI, Streamlit, Gradio, .NET, ASP.NET Core, SQL Server
- **Tools:** Git, Github, VS Code, Jupyter Notebook, Google Colab, N8N, Kali Linux, Postman

ACHIEVEMENTS AND PROBLEM SOLVING

I have solved **2600+** **Data Structure and Algorithm** problems across various online judges and Participated in 100+ Online Programming contests.

- **Codeforces Profiles (Current and Previous):** Max rating 1497, Problems solved 1580 (Combined)
- **Leetcode Profile:** Max rating 1739, Problems solved 173
- **Codechef Profile:** Max rating 1678, Problems solved 343
- **Vjudge Profile:** Problems solved 352
- **CSes Profile:** Problems solved 85
- **National Level IUPC Participation (Team: JUST BitWarriors):** PSTU IUPC South Zone 2026, Rank **16/41** | IUT IUPC 2026, Rank **61/112** | DUET IUPC 2026, Rank **94/137**
- **Champion** of JUST intra University Programming Contest 2025

PROJECTS

Smart Aquaculture: Edge-AI & Autonomous IoT Pond System | C++, ESP32, TinyML, Python, Next.js, Node.js

- Designed an autonomous pond monitoring and control node on an ESP32 microcontroller interfaced with pH, turbidity (ADC), DS18B20 (1-Wire digital), and rain sensors.
- Implemented on-device Digital Signal Processing (DSP) via an Exponential Moving Average (EMA) low-pass filter ($\alpha = 0.25$) to suppress analog voltage noise and motor back-EMF spikes.
- Engineered a TinyML synthesis pipeline compiling a scikit-learn decision tree classifier into zero-heap, inline C++ firmware (< 1 ms inference latency, < 1 KB flash footprint).
- Programmed autonomous fail-safe routines triggering emergency aeration via active-low relay isolation directly at the hardware edge during network blackouts.
- Integrated scheduled feeding via PWM-controlled MG 996R servo gate and telemetry transmission to a full-stack dashboard.

[Github Repository](#)

Hotel Management System | ASP.NET Core (.NET 10), React 19, SQL Server, EF Core

- Built a full stack hotel management system with real time analytics, booking lifecycle management, role based access, and payment tracking using Clean Architecture (Domain, Application, Infrastructure, API)
- Implemented booking flow across Pending, Confirmed, Checked In, and Checked Out stages with JWT authentication, BCrypt hashing, and Admin/Staff role management
- Delivered full CRUD for Rooms, Customers, Employees, Bookings, and Payments, and validated the system with xUnit tests using EF Core InMemory

[Github Repository](#)

Insurance Premium Predictor | Python, FastAPI, Scikit-learn, Pandas, Matplotlib, Streamlit

- Developed and deployed a full-stack ML web application to predict health and car insurance categories.
- Built a FastAPI-based inference backend and integrated with a Streamlit frontend for real-time predictions and analytics.

[Github Repository](#), [Live Demo](#)

ADDITIONAL SKILLS

Languages: English (Fluent), Bengali (Native), Hindi (Conversational)

Soft Skills: Analytical thinking, Team collaboration, Adaptability, communication

Core Competencies: DSA, OOP, OS, Networking, DBMS, IOT and Embedded systems, AI and Machine Learning